

1. Give a context-free grammar (CFG) for each of the following languages. You must provide an explanation for how your grammar works by describing in English what is generated by each non-terminal. (Formal proofs of correctness are not required.)

(a) $\{x(110)^n x^R : x \in \{0,1\}^*, n \geq 1\}$

Solution:

$$S \rightarrow OS0 \mid 1S1 \mid A$$

$$A \rightarrow 110A \mid 110$$

Explanation:

- A generates strings of the form $(110)^n$ for $n \geq 1$.
- S generates palindromes by wrapping matching 0s and 1s around strings generated by A .
- Therefore, S generates all strings of the form $x(110)^n x^R$ for $x \in \{0,1\}^*$ and $n \geq 1$.

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(b) $\{1^i 0^j 1^k : j = 2i + 3k, i, j, k \geq 0\}$

Solution:

$$S \rightarrow AB$$

$$A \rightarrow 1A00 \mid \varepsilon$$

$$B \rightarrow 000B1 \mid \varepsilon$$

Explanation:

- A generates strings of the form $1^i 0^{2i}$ for $i \geq 0$.
- B generates strings of the form $0^{3k} 1^k$ for $k \geq 0$.
- Hence, S produces strings of the form $1^i 0^j 1^k$ where $j = 2i + 3k$.

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(c) $\{1^i 0^j 1^k : i + k \equiv 0 \pmod{3}, 0 \leq j \leq k\}$

Solution:

$$\begin{aligned}
 S &\rightarrow A_0 B_0 A_0 \mid A_0 B_1 A_2 \mid A_0 B_2 A_1 \\
 &\mid A_1 B_0 A_2 \mid A_1 B_1 A_1 \mid A_1 B_2 A_0 \\
 &\mid A_2 B_0 A_1 \mid A_2 B_1 A_0 \mid A_2 B_2 A_2 \\
 A_0 &\rightarrow 111A_0 \mid \varepsilon \\
 A_1 &\rightarrow 1A_0 \\
 A_2 &\rightarrow 11A_0 \\
 B_0 &\rightarrow 000B_0 111 \mid \varepsilon \\
 B_1 &\rightarrow 0B_0 1 \\
 B_2 &\rightarrow 00B_0 11
 \end{aligned}$$

Explanation:

- A_p generates all 1^i where $i \equiv p \pmod{3}$.
- B_q generates all $0^j 1^j$ where $j \equiv q \pmod{3}$ (so $j \leq k$ is guaranteed).
- Each production $A_p B_q A_r$ ensures $i + j + k \equiv 0 \pmod{3}$ with $i \equiv p$, $k \equiv r$, and $j \equiv q$.
- Therefore, S generates all strings $1^i 0^j 1^k$ where $i + k \equiv 0 \pmod{3}$ and $j \leq k$.

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2. For each of the following languages, decide whether it is regular, irregular, context-free, not-context free or some combination.

- (a) Let $\Sigma = \{0, 1, 2\}$ and define the language

$$L_1 = \{0^n w 1^n : w \in \{2\}^*, n \geq 0\}$$

Solution: Not regular. Context-free.

To show L_1 is not regular, we use a fooling set. Let $F = \{0^n : n \geq 0\}$. Let $x = 0^i$ and $y = 0^j$ be arbitrary strings in F with $i \neq j$.

Let $z = 2^k 1^i$ for any $k \geq 0$. Then:

$$xz = 0^i 2^k 1^i \in L_1 \quad (\text{since the number of 0s and 1s match}),$$

$$yz = 0^j 2^k 1^i \notin L_1 \quad (\text{since } j \neq i).$$

Thus, F is a fooling set for L_1 . Since F is infinite, L_1 is not regular.

We now give a context-free grammar for L_1 :

$$S \rightarrow 0S1 \mid T, \quad T \rightarrow 2T \mid \varepsilon$$

This grammar ensures equal numbers of 0s and 1s are placed around any number of 2s in the middle. Hence, L_1 is context-free. ■

- (b) Let $\Sigma = \{0, 1\}$ and define the language:

$$L_2 = \{x0y : x, y \in 1^*, \text{ and } |x| > 0, |y| > 2\}$$

Solution: Regular. Context-free.

This language is regular because you can form a regular expression that describes it:

$$1^+ 0 111^+ \tag{1}$$

Because the language is regular, it is also context-free. ■